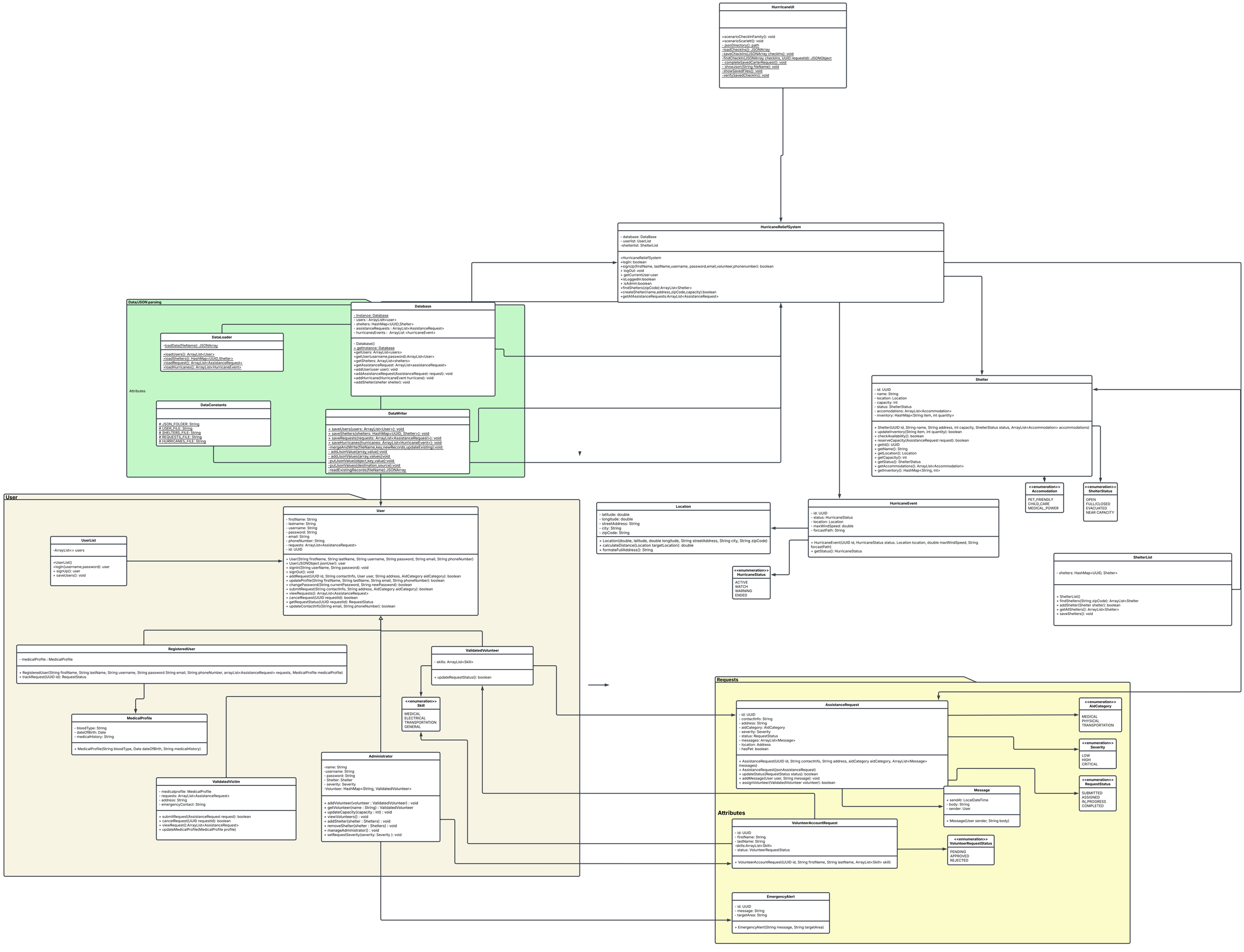

HurrricaneUI
+scenarioCheckInFamily(): void
+scenarioScarlett(): void
- jsonDirectory(): path
-loadCheckIns(): JSONArray
-saveCheckIns(JSONArray checkIns): void
-findCheckIn(JSONArray checkins, UUID requestId): JSONObject
- completeSavedCarterRequest(): void
- showJson(String fileName): void
-showSavedFiles(): void
-verifySavedCheckIn(): void
HurricaneReliefSystem
- database: DataBase
- userlist: UserList
-shelterlist: ShelterList
+HurricaneReliefSystem
+logIn: boolean
+signUp(firstName, lastName,username, password,email,volunteer,phonenumber): boolean
+ logOut: void
+ getCurrentUser:user
+isLoggedIn:boolean
+ isAdmin:boolean
+findShelters(zipCode):ArrayList<Shelter>
+createShelter(name,address,zipCode,capacity):boolean
+getAllAssistanceRequests:ArrayList<AssistanceRequest>
Data/JSON parsing
Database
- Instance: Database
- users : ArrayList<user>
- shelters: HashMap<UUID,Shelter>
- assistanceRequests : ArrayList<AssistanceRequest>
- hurricanesEvents : ArrayList <hurricaneEvent>
- Database()
+ getInstance: Database
+getUsers: ArrayList<users>
+getUser(username,password):ArrayList<User>
+getShelters: ArrayList<shelters>
+getAssistanceRequest: ArrayList<assistanceRequest>
+addUser(user user): void
+addAssistanceRequest(AssistanceRequest request): void
+addHurricane(HurricaneEvent hurricane): void
+addShelter(shelter shelter): void
DataLoader
-loadData(fileName): JSONArray
+loadUsers(): ArrayList<User>
+loadShelters(): HashMap<UUID,Shelter>
+loadRequest(): ArrayList<AssistanceRequest>
+loadHurricanes(): ArrayList<HurricaneEvent>
Attributes
DataConstants
JSON_FOLDER: String
USER_FILE: String
SHELTERS_FILE: String
REQUESTS_FILE: String
HURRICANES_FILE: String
DataWriter
+ saveUsers(users: ArrayList<User>): void
+ saveShelters(shelters: HashMap<UUID, Shelter>): void
+ saveRequests(requests: ArrayList<AssistanceRequest>): void
+ saveHurricanes(hurricanes: ArrayList<HurricaneEvent>): void
-mergeAndWrite(fileName,key,newRecords,updateExisting):void
- addJsonValue(array,value):void
- addJsonValues(array,values):void
-putJsonValue(object,key,value):void
-putJsonValues(destination,source):void
-readExistingRecords(fileName):JSONArray
Shelter
- id: UUID
- name: String
- location: Location
- capacity: int
- status: ShelterStatus
- accomodations: ArrayList<Accommodation>
- inventory: HashMap<String item, int quantity>
+ Shelter(UUID id, String name, String address, int capacity, ShelterStatus status, ArrayList<Accommodation> accommodations)
+ updateInventory(String item, int quantity): boolean
+ checkAvailability(): boolean
+ reserveCapacity(AssistanceRequest request): boolean
+ getId(): UUID
+ getName(): String
+ getLocation(): Location
+ getCapacity(): int
+ getStatus(): ShelterStatus
+ getAccommodations(): ArrayList<Accommodation>
+ getInventory(): HashMap<String, int>
<<enumeration>>
Accomodation
PET_FRIENDLY
CHILD_CARE
MEDICAL_POWER
<<enumeration>>
ShelterStatus
OPEN
FULL/CLOSED
EVACUATED
NEAR CAPACITY
User
UserList
-ArrayList<> users
+UserList()
+loginUser(username,password): user
+ signUp(): user
+ saveUsers(): void
User
- firstName: String
- lastname: String
- username: String
- password: String
- email: String
- phoneNumber: String
- requests: ArrayList<AssistanceRequest>
- id: UUID
+ User(String firstName, String lastName, String username, String password, String email, String phoneNumber)
+ User(JSONObject jsonUser): user
+ signIn(String userName, String password): void
+ signOut(): void
+ addRequest(UUID id, String contactInfo, User user, String address, AidCategory aidCategory): boolean
+ updateProfile(String firstName, String lastName, String email, String phoneNumber): boolean
+ changePassword(String currentPassword, String newPassword): boolean
+ submitRequest(String contactInfo, String address, AidCategory aidCategory): boolean
+ viewRequests(): ArrayList<AssistanceRequest>
+ cancelRequest(UUID requestId): boolean
+ getRequestStatus(UUID requestId): RequestStatus
+ updateContactInfo(String email, String phoneNumber): boolean
Location
- latitude: double
- longitude: double
- streetAddress: String
- city: String
- zipCode: String
+ Location(double, latitude, double longitude, String streetAddress, String city, String zipCode)
+ calculateDistance(Location targetLocation): double
+ formateFullAddress(): String
HurricaneEvent
- id: UUID
- status: HurricaneStatus
- location: Location
- maxWindSpeed: double
- forcastPath: String
+ HurricaneEvent(UUID id, HurricaneStatus status. Location location, double maxWindSpeed, String forcastPath)
+ getStatus(): HurricaneStatus
<<enummeration>>
HurricaneStatus
ACTIVE
WATCH
WARNING
ENDED
ShelterList
- shelters: HashMap<UUID, Shelter>
+ ShelterList()
+ findShelters(String zipCode): ArrayList<Shelter
+ addShelter(Shelter shelter): boolean
+ getAllShelters(): ArrayList<Shelter>
+ saveShelters(): void
RegisteredUser
- medicalProfile : MedicalProfile
+ RegisteredUser(String firstName, String lastName, String username, String password String email, String phoneNumber, arrayList<AssistanceRequest> requests, MedicalProfile medicalProfile)
+ trackRequest(UUID id): RequestStatus
ValidatedVolunteer
- skills: ArrayList<Skill>
+ updateRequestStatus(): boolean
MedicalProfile
- bloodType: String
- dateOfBirth: Date
- medicalHistory: String
+ MedicalProfile(String bloodType, Date dateOfBirth, String medicalHistory)
<<enumeration>>
Skill
MEDICAL
ELECTRICAL
TRANSPORTATION
GENERAL
Administrator
-name: String
-username: String
- password: String
- Shelter: Shelter
- severity: Severity
-Volunteer: HashMap<String, ValidatedVolunteer>
+ addVolunteer(volunteer : ValidatedVolunteer) : void
+ getVolunteer(name : String) : ValidatedVolunteer
+ updateCapacity(capacity : int) : void
+ viewVolunteers() : void
+ addShelter(shelter : Shelters) : void
+ removeShelter(shelter : Shelters) : void
+ manageAdministrator() : void
+ setRequestSeverity(severity: Severity): void
ValidatedVictim
- medicalprofile: MedicalProfile
- requests: ArrayList<AssistanceRequest>
- address: String
- emergencyContact: String
+ submitRequest(AssistanceRequest request): boolean
+ cancelRequest(UUID requestId): boolean
+ viewRequest():ArrayList<AssistanceRequest>
+ updateMedicalProfile(MedicalProfile profile)
Requests
AssistanceRequest
- id: UUID
- contactInfo: String
- address: String
- aidCategory: AidCategory
- severity: Severity
- status: RequestStatus
- messages: ArrayList<Message>
- location: Address
- hasPet: boolean
+ AssistanceRequest(UUID id, String contactInfo, String address, aidCategory aidCategory, ArrayList<Message> messages)
+ AssistanceRequest(jsonAssistanceRequest)
+ updateStatus(RequestStatus status): boolean
+ addMessage(User user, String message): void
+ assignVolunteer(ValidatedVolunteer volunteer): boolean
<<enumeration>>
AidCategory
MEDICAL
PHYSICAL
TRANSPORTATION
<<enumeration>>
Severity
LOW
HIGH
CRITICAL
<<enumeration>>
RequestStatus
SUBMITTED
ASSIGNED
IN_PROGRESS
COMPLETED
Message
+ sendAt: LocalDateTime
- body: String
- sender: User
+ Message(User sender, String body)
Attributes
VolunteerAccountRequest
- id: UUID
- firstName: String
- lastName: String
-skills:ArrayList<Skill>
- status: VolunteerRequestStatus
+ VolunteerAccountRequest(UUID id, String firstName, String lastName, ArrayList<Skill> skill)
<<ennumeration>>
VolunteerRequestStatus
PENDING
APPROVED
REJECTED
EmergencyAlert
- id: UUID
- message: String
- targetArea: String
+ EmergencyAlert(String message, String targetArea)